

DATA LANDSCAPE ASSIGNMENT

WEB SCRAPING

1. Description of the Website Scraped

Books to Scrape (<https://books.toscrape.com/>) is a demo site created specifically for practicing web scraping. It lists books with prices, ratings, availability, and detail pages. This makes it ideal for learning and demonstrations without violating site policies.

2. Fields Extracted

From list and detail pages:

- title – Book title
- price – Original text price (e.g., £51.77)
- price_numeric – Numeric value parsed from price (e.g., 51.77)
- rating – Star rating mapped to 1–5
- availability – Availability text from listing
- product_page_url – Absolute URL to the book detail page
- upc – Unique product code from detail table
- category – Category from breadcrumb
- description – Book description (if available)

3. Security Approach Used

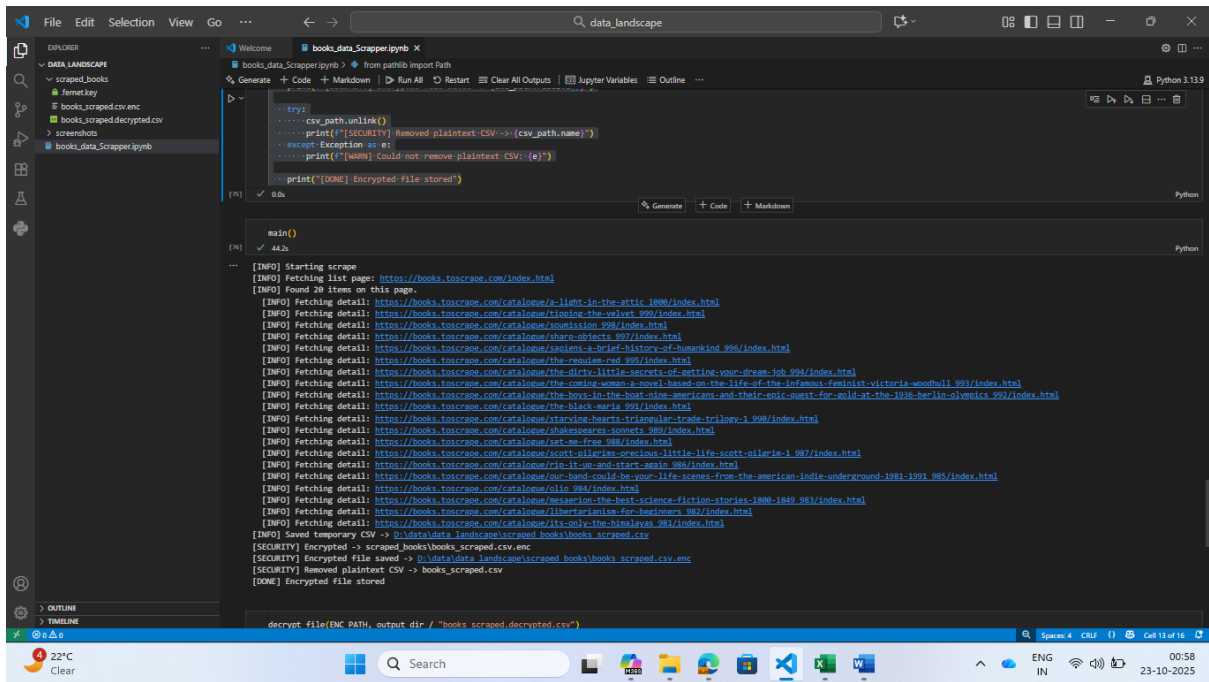
- **At-rest encryption with Fernet:**
 - The scraped dataset is first saved as a temporary CSV.
 - The CSV is encrypted with Fernet (symmetric encryption ensuring confidentiality and integrity).
 - The plaintext CSV is deleted, leaving only the .enc file.
- **Key management:**
 - The script uses FERNET_KEY if set in the environment.
 - If not provided, it generates and stores a key at scraped_books/.fernet.key and restricts permissions where possible.

4. Challenges Faced

Encoding issues: Pound symbol £ sometimes appears as Â£ due to encoding misdetection.

Fix: Force resp.encoding = 'utf-8' and use regex to extract numeric price (r'(\d+(?:\.\d+)?)')

5. Screenshots and Code Snippets

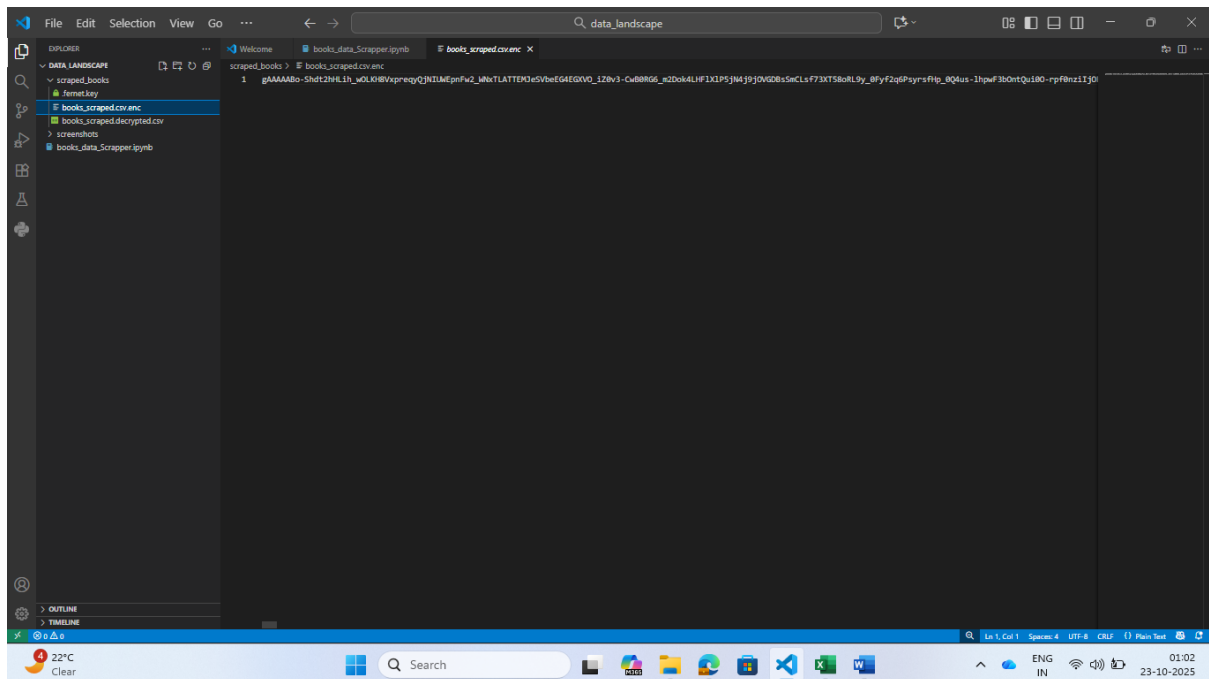


```
try:
    csv_path.unlink()
except:
    print("[SECURITY] Removed plaintext CSV -> {csv_path.name}")
except Exception as e:
    print(f"[WARN] Could not remove plaintext CSV: {e}")

print("[DONE] Encrypted file stored")

def main():
    [INFO] Starting scrape
    [INFO] Fetching list page: https://books.toscrape.com/index.html
    [INFO] Found 28 items on this page.
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/a-light-in-the-attic-1000/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/ringing-the-volcani-999/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/commission-999/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/sharp-objects-997/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/specimens-a-brief-history-of-humanind-996/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/the-requiem-red-995/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/the-divys-little-search-for-petting-your-dream-job-994/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/the-coming-woman-a-novel-based-on-the-life-of-the-infamous-feminist-victoria-woodhull-993/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/the-black-man-991/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/starving-heart-triangular-trade-trilogy-1-990/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/shakespeares-sonnets-989/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/set-me-free-988/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/scott-pilgrims-peculiar-little-life-scott-pilgrim-1-987/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/rip-it-up-and-start-again-986/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/our-band-could-be-your-life-scenes-from-the-american-indie-underground-1981-1991-985/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/olio-984/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/stepped-heaven-1889-1849-983/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/libertarianism-for-beginners-982/index.html
    [INFO] Fetching detail: https://books.toscrape.com/catalogue/its-only-the-himalayas-981/index.html
    [INFO] Saved temporary CSV -> D:\data\data-landscape\scraped-books\books-scraped.csv
    [SECURITY] Encrypted -> scraped_books\books_scraped.csv.enc
    [SECURITY] Removed plaintext CSV -> books_scraped.csv
    [DONE] Encrypted file stored
```

Figure1 Extracting Data



```
scraped_books -> books_scraped.csv.enc
1 gAAAAABo-Shdt2HhLh_wdX0BvXpreyQJNIMEpfnw2_WNtLATtDheSVbeEGEGOV0_12v3-CuB8G6_m2Dok4HfX1P5jM4j9J0V020B4ScLsf7XTS80L8y_8fy72qPyrsf8p_0Q0us-1hpuf30ntQu180-rp8enz1j0
```

Figure 2 Extracted Data (Encrypted)

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. [Don't show again](#) [Save As...](#)

	A	B	C	D	E	F	G	H	I
	title	price	rating	availability	product_page_url	upc	category	description	price_numeric
1	A Light in t	£51.77		3 In stock	https://books.toscrape.com/catalogue/a897fe39b1053632		Poetry	It's hard to imagine a world wit	51.77
2	Tipping th	£53.74		1 In stock	https://books.toscrape.com/catalogue/90fa61229261140a		Historical Fiction	"Erotic and absorbing...Written	53.74
3	Soumissic	£50.10		1 In stock	https://books.toscrape.com/catalogue/6957f44c3847a760		Fiction	Dans une France assez proche	50.1
4	Sharp Obj	£47.82		4 In stock	https://books.toscrape.com/catalogue/e00eb4fd7b871a48		Mystery	WICKED above her hipbone, Gif	47.82
5	Sapiens: /	£54.23		5 In stock	https://books.toscrape.com/catalogue/4165285e1663650f		History	From a renowned historian cor	54.23
6	The Requi	£22.65		1 In stock	https://books.toscrape.com/catalogue/f77dbf2323deb740		Young Adult	Patient Twenty-nine. A monster	22.65
7	The Dirty	£33.34		4 In stock	https://books.toscrape.com/catalogue/2597b5a345f45e1b		Business	Drawing on his extensive experi	33.34
8	The Comin	£17.93		3 In stock	https://books.toscrape.com/catalogue/e72a5dfc7e9267b2		Default	"If you have a heart, if you have	17.93
9	The Boys	£22.60		4 In stock	https://books.toscrape.com/catalogue/e10e1e165dc8be4a		Default	For readers of Laura Hillenbrar	22.6
10	The Black	£52.15		1 In stock	https://books.toscrape.com/catalogue/1dfe412b8ac00530		Poetry	Praise for Aracelis Girmay: "Gir	52.15
11	Starving H	£13.99		2 In stock	https://books.toscrape.com/catalogue/0312262ecafa5a40		Default	Since her assault, Miss Annette	13.99
12	Shakespeare	£20.66		4 In stock	https://books.toscrape.com/catalogue/30a7f60cd76ca58c		Poetry	This book is an important and	20.66
13	Set Me Fre	£17.46		5 In stock	https://books.toscrape.com/catalogue/c66396b0f23f6ecc		Young Adult	Aaron Ledbetterâ€s future hac	17.46
14	Scott Pilgr	£52.29		5 In stock	https://books.toscrape.com/catalogue/3b1c02bac2a429e6		Sequential Art	Scott Pilgrim's life is totally swe	52.29
15	Rip it Up a	£35.02		5 In stock	https://books.toscrape.com/catalogue/a34ba96d4081e6a4		Music	Punk's raw power rejuvenated i	35.02
16	Our Band	£57.25		3 In stock	https://books.toscrape.com/catalogue/deda3e61b9514b83		Music	This is the never-before-told stc	57.25
17	Olio	£23.88		1 In stock	https://books.toscrape.com/catalogue/feb7cc7701ecf901		Poetry	Part fact, part fiction, Tyeihlmbe	23.88
18	Mesaerio	£37.59		1 In stock	https://books.toscrape.com/catalogue/e30f54cea9b38190		Science Fiction	Andrew Barger, award-winning	37.59
19	Libertaria	£51.33		2 In stock	https://books.toscrape.com/catalogue/a18a4f574854aced		Politics	Libertarianism isn't about winn	51.33

Figure 3 Decrypted Data

6. Ethical Considerations

- **Purpose-built site:** The project uses a site designed for scraping practice.
- **Respectful access pattern:** Short delays (1–2s), retries with backoff, and request timeouts.
- **Data handling:** Demonstrates secure storage (encryption at rest) and avoids distribution of plaintext datasets.